

Predicting Return of Spontaneous Circulation in Out-of-Hospital Cardiac Arrest

A NEMSIS 2025 analysis of EMS system response time, clinical context, and Utstein-style cohorts

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AI Mastery Capstone - Project 4

17,198	1,777	0.7321
Broad cohort records	Classic Utstein records	Best broad ROC AUC

Research question: Does EMS system response time add predictive value for ROSC beyond age, geography, arrest etiology, witness category, pre-arrival AED use, and initial rhythm?

Data source: National Highway Traffic Safety Administration, National Emergency Medical Services Information System (NEMSIS). Derived content remains NHTSA property. NHTSA is not responsible for claims arising from this analysis.

This report describes predictive associations. It does not estimate a causal effect of response time and does not report survival to hospital discharge.

Executive Summary

This project used a supervised binary-classification workflow to predict any return of spontaneous circulation (ROSC) among out-of-hospital cardiac-arrest EMS activations. A 50,000-record cardiac-arrest sample was linked to NEMSIS computed response-time, resuscitation, ROSC, and witness tables without extracting the 43 GB primary events file.

- The final broad modeling cohort contained 17,198 first-unit, primary-area emergency responses with attempted resuscitation, known ROSC, and recorded response time.
- The classic presumed-cardiac Utstein-style subgroup contained 1,777 bystander-witnessed arrests with an initially shockable rhythm.
- ROSC occurred in 29.64% of the broad cohort and 53.91% of the classic Utstein subgroup.
- Within the classic subgroup, ROSC declined from 62.92% in the fastest response-time decile to 43.68% in the slowest decile.
- Adding response time improved cross-validated ROC AUC for both logistic regression and random forest in both cohorts.
- The enhanced random forest produced the highest threshold-independent performance: ROC AUC 0.7321 in the broad cohort and 0.6095 in the classic subgroup.

Primary conclusion: response time contributed consistent but modest incremental predictive value. Initial rhythm and other arrest characteristics remained central. The results support including response time in cardiac-arrest outcome models while avoiding causal or clinical-decision claims.

Project Design

The project was designed around two nested analyses:

Cohort	Purpose	Records	ROSC rate
Broad	Develop and compare general ROSC classifiers	17,198	29.64%
Utstein-style	Bystander-witnessed and shockable rhythm	2,054	53.21%
Classic Utstein	Adds presumed cardiac etiology	1,777	53.91%

The classic subgroup is described as Utstein-style ROSC analysis, not Utstein survival. The available target is any ROSC recorded during EMS care; survival to hospital discharge was not established.

1. Data Source and Cohort Construction

Data source

The analysis used the 2025 NEMSIS Public-Release Research Dataset, which contains 63,635,893 EMS activations submitted by 14,801 agencies serving 54 states and territories (National Emergency Medical Services Information System [NEMSIS], 2026). NEMSIS describes the data as a de-identified convenience sample rather than a population-based dataset. Source tables were streamed directly from the compressed release to limit storage and preserve the original archive.

Cohort flow

Stage	Eligibility rule	Records
Starting cardiac-arrest sample	Cardiac arrest before EMS arrival	50,000
Attempted resuscitation	Defibrillation, ventilation, or chest compressions attempted	34,773
Known ROSC	Unambiguous positive or negative eArrest.12 outcome	33,608
Response time recorded	EMSSystemResponseTimeMin present	33,574
First EMS unit	eScene.01 = Yes	17,802
Primary emergency response	eResponse.05 = Primary Response Area	17,198
Classic Utstein	Clean bystander witness + shockable rhythm + presumed cardiac	1,777

Outcome definition

ROSC was coded positive when eArrest.12 contained Yes at ED arrival, Yes prior to ED arrival, or Yes sustained for 20 consecutive minutes. It was coded negative when eArrest.12 contained No. Records containing both positive and negative codes, or only missing/not-applicable codes, were excluded from outcome-eligible cohorts (NEMSIS, 2024).

Response-time definition

EMS system response time is the interval from unit notification by dispatch (eTimes.03) to unit arrival on scene (eTimes.06). Zero-minute values were retained because they are permitted by the NEMSIS computed-variable range and may represent identical recorded timestamps. Three broad-cohort records exceeded 60 minutes (64.3, 69.0, and 153.02 minutes); they were retained because the values were internally computable and within the NEMSIS public-release bounds. The classic subgroup maximum was 55 minutes (NEMSIS, 2024).

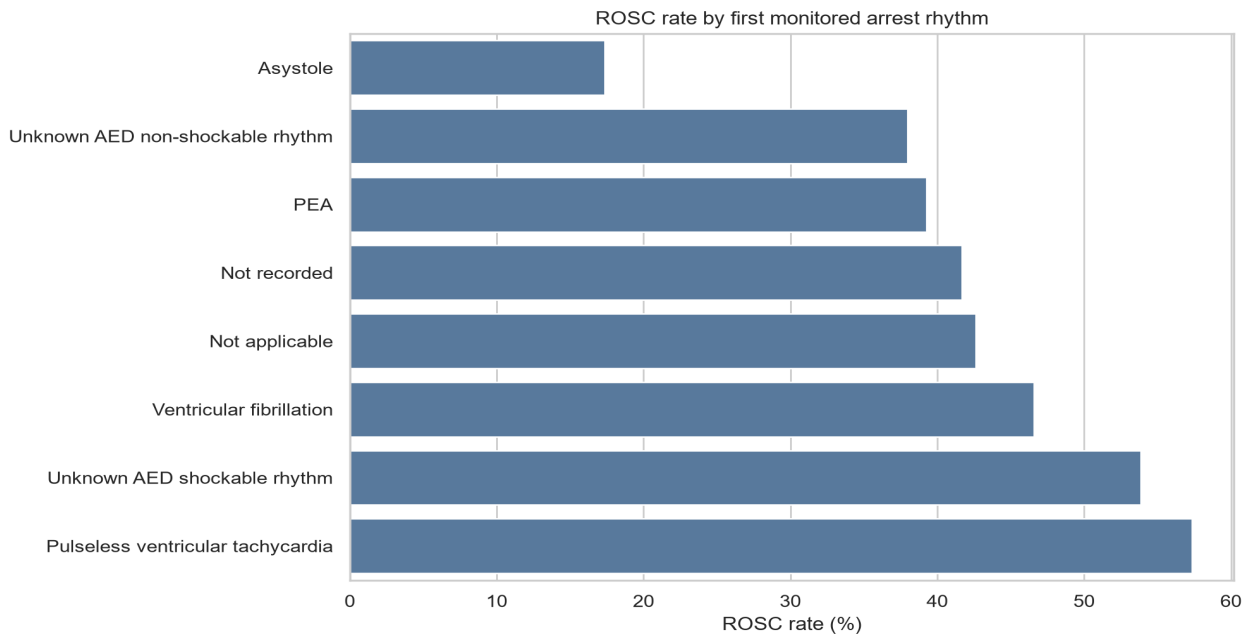
2. Variables and Preprocessing

Role	Variables	Processing
Outcome	ROSC	Binary 0/1 target from eArrest.12
Numeric	Age; EMS system response time	Median imputation; standardized for logistic regression
Categorical	Urbanicity; Census region; etiology; witness; AED; rhythm	Explicit missing categories and one-hot encoding
Cohort flags	Bystander witnessed; shockable rhythm; Utstein-style; classic Utstein	Rule-based indicators; not used as predictors

The local modeling file omitted PcrKey and contained only the 17,198 broad-cohort rows. The row-level CSV was excluded from Git because NEMSIS terms restrict redistribution. Reproducible preparation and modeling code, aggregate results, and figures were retained in the project repository.

Initial rhythm

Initial rhythm showed the strongest descriptive separation among available variables. Asystole had a 17.34% ROSC rate, compared with 46.57% for ventricular fibrillation and 57.34% for pulseless ventricular tachycardia. This justified including rhythm in the baseline model before testing response time's incremental value.

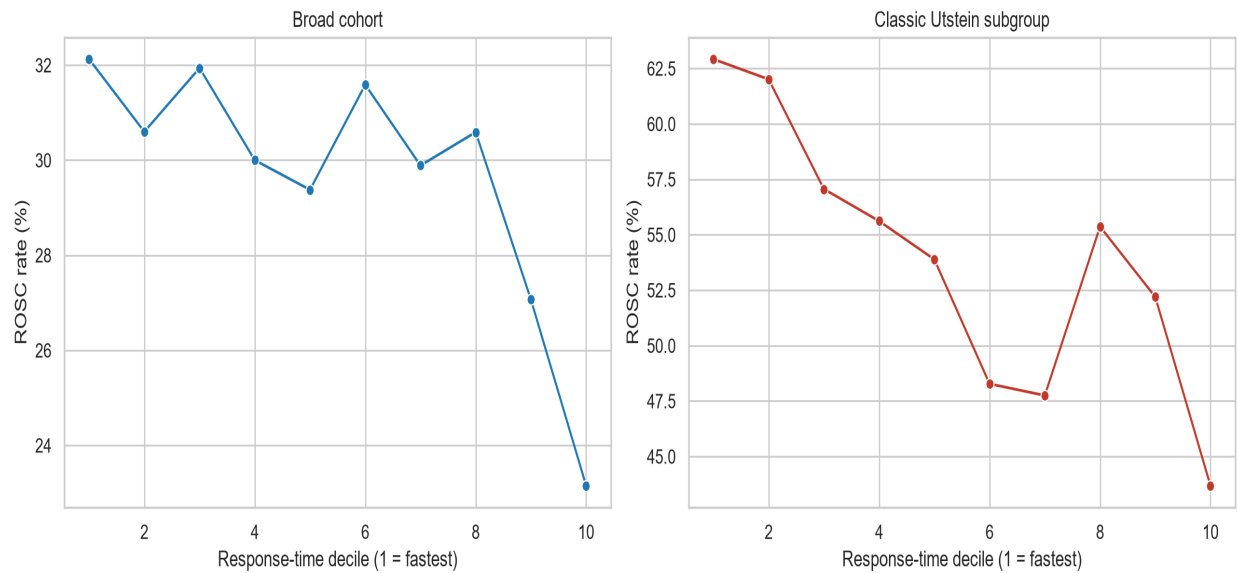


AED use also separated outcomes: pre-arrival AED defibrillation was associated with 44.58% ROSC versus 29.17% where no AED was used. These are descriptive associations and may reflect underlying rhythm and witness differences.

3. Response-Time Pattern in the Classic Utstein Subgroup

The classic subgroup provided a clinically familiar, more homogeneous comparison: presumed-cardiac, bystander-witnessed arrest with an initial shockable rhythm. This comparator follows the standardized Utstein reporting framework (Bray et al., 2024). Its 1,777 records were divided into equal-frequency response-time deciles.

ROSC rate by EMS system response-time decile



Decile	n	Median min	Max min	ROSC
1	178	2.06	2.97	62.92%
2	179	3.40	3.75	62.01%
3	177	4.02	4.30	57.06%
4	178	4.73	5.02	55.62%
5	180	5.42	5.82	53.89%
6	174	6.03	6.45	48.28%
7	178	6.98	7.53	47.75%
8	177	8.32	9.17	55.37%
9	182	10.23	12.00	52.20%
10	174	14.66	55.00	43.68%

The overall gradient is downward but not perfectly monotonic. Deciles 8 and 9 rebound relative to deciles 6 and 7, reinforcing the decision to evaluate a nonlinear model and avoid interpreting the descriptive pattern as a dose-response causal effect.

4. Modeling Methods

Algorithms

- Logistic regression served as the interpretable linear baseline.
- Random forest used 300 trees, maximum depth 12, minimum leaf size 10, square-root feature sampling, and balanced-subsample class weighting.

Model comparison

For each algorithm and cohort, a baseline model used age, urbanicity, Census region, arrest etiology, witness category, pre-arrival AED use, and initial rhythm. The enhanced model added EMS system response time. Five-fold stratified cross-validation with a fixed random seed generated out-of-fold probabilities for every record.

Evaluation Metrics Justification

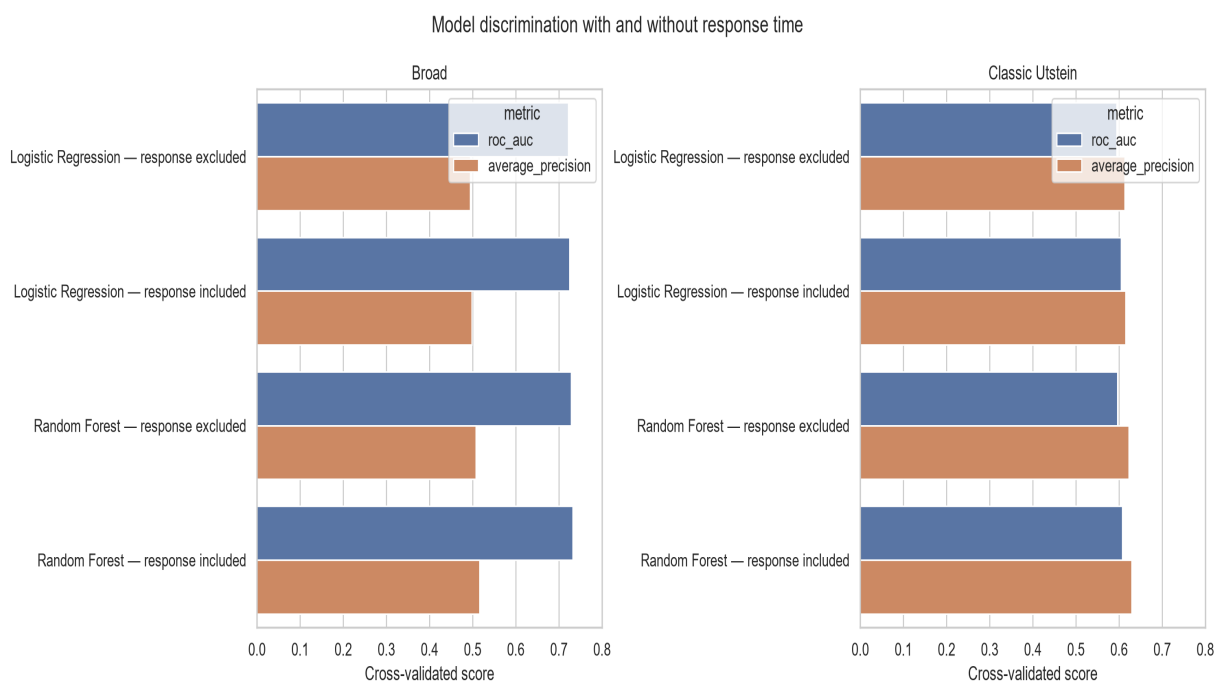
Task and model fit: This project is a supervised binary classification task that predicts whether return of spontaneous circulation (ROSC) was documented for each NEMSIS cardiac-arrest record. Logistic regression and random forest both produce class probabilities, so discrimination and positive-case retrieval metrics are appropriate for comparing the two model types.

ROC AUC: ROC AUC measures how well each model ranks ROSC-positive records above ROSC-negative records across all possible thresholds instead of depending on one cutoff. This is appropriate for comparing logistic regression and random forest even though the models can produce differently distributed probabilities.

Average precision and class imbalance: The broad cohort contains 5,097 ROSC-positive records among 17,198 records, a positive rate of 29.64%. Because negative outcomes are the majority, ordinary accuracy can appear acceptable even when a model performs poorly on the ROSC-positive class. Average precision is therefore a primary metric because it summarizes precision-recall performance for the less common positive outcome and can be interpreted relative to the 0.2964 positive-class prevalence baseline (Saito & Rehmsmeier, 2015).

Balanced accuracy and F1: Balanced accuracy gives equal importance to sensitivity for ROSC-positive records and specificity for ROSC-negative records. F1 summarizes the balance between precision and recall at the fixed 0.50 threshold. This matters because false positives overstate ROSC likelihood while false negatives miss positive outcomes. These threshold-dependent metrics complement, rather than replace, threshold-independent ROC AUC and average precision.

Validation design: Five-fold stratified cross-validation preserves the ROSC class proportion within every fold and generates an out-of-fold prediction for every record. This provides a more stable comparison than one train-test split and reduces the risk that a favorable or unfavorable single split determines the reported performance.



5. Model Results

Cohort	Algorithm	Response time	ROC AUC	Avg. precision	Accuracy	Bal. accuracy	F1
Broad	Logistic	No	0.7215	0.4946	0.7193	0.5891	0.3624
Broad	Logistic	Yes	0.7241	0.4979	0.7218	0.5959	0.3793
Broad	Random forest	No	0.7289	0.5088	0.6554	0.6787	0.5587
Broad	Random forest	Yes	0.7321	0.5160	0.6591	0.6805	0.5604
Classic	Logistic	No	0.5947	0.6129	0.5774	0.5682	0.6363
Classic	Logistic	Yes	0.6049	0.6154	0.5853	0.5758	0.6441
Classic	Random forest	No	0.5933	0.6199	0.5689	0.5691	0.5864
Classic	Random forest	Yes	0.6095	0.6328	0.5937	0.5921	0.6192

Broad cohort

Adding response time increased logistic ROC AUC by 0.0026 and random-forest ROC AUC by 0.0032. The enhanced random forest achieved the strongest broad-cohort discrimination (ROC AUC 0.7321; average precision 0.5160). Its raw accuracy was lower than logistic accuracy because balanced-subsample weighting increased sensitivity to the minority ROSC-positive class. Its balanced accuracy (0.6805) and F1 (0.5604) were substantially higher than the enhanced logistic values.

Classic Utstein subgroup

Response time produced a larger incremental gain in the standardized subgroup. Logistic ROC AUC increased by 0.0102, and random-forest ROC AUC increased by 0.0162. The enhanced random forest also improved average precision by 0.0129 and F1 by 0.0328 compared with the random-forest baseline. The enhanced logistic model retained the highest classic-subgroup F1 at the fixed 0.50 threshold, illustrating that model preference depends on the operational metric.

Selection: the enhanced random forest was selected as the leading classifier based on ROC AUC and average precision. Logistic regression remained an important interpretability benchmark.

6. Interpretation and Operational Relevance

The findings align with the department's use of Utstein-style cardiac-arrest review. The standardized subgroup made the response-time gradient more visible and increased the incremental model value of response time. This does not mean the subgroup model was more accurate overall; its smaller sample and reduced case-mix variation limited discrimination. Instead, the subgroup is useful for comparing a more clinically consistent set of arrests.

- Use the broad model for general predictive benchmarking and feature comparison.
- Use the classic Utstein analysis for a recognizable quality-review lens.
- Report response time as one factor among rhythm, witness status, AED use, etiology, age, and geography.
- Do not use this model for individual patient treatment, dispatch prioritization, or causal performance claims.

Bias and Responsible Use

Potential sources of bias: NEMSIS is a de-identified convenience sample rather than a nationally representative patient sample. Participation, case mix, response systems, and documentation practices can differ across states, agencies, Census regions, and urbanicity categories. Missing and not-recorded values may reflect agency workflow rather than the patient condition. These differences can introduce representation bias and measurement bias into both training and evaluation.

Possible real-world impact: The model may perform better for well-represented agencies or urban systems and may understate or overstate ROSC probability for rural, frontier, regional, or age subgroups. If such predictions were used for treatment or dispatch decisions, unequal error rates could reinforce existing disparities. ROSC is also only an intermediate outcome, so it must not be treated as a substitute for survival to discharge or favorable neurological outcome.

Proposed mitigation step: Before any operational use, the model should be externally validated with data from separate agencies and audited by urbanicity, Census region, and age band using ROC AUC, average precision, balanced accuracy, recall, and calibration. Material subgroup gaps should trigger investigation, collection of more representative data, and reweighting or retraining; use should be withheld for groups that do not demonstrate acceptable performance. Until those checks are completed, this model remains an educational quality-improvement prototype and must not be used for individual patient treatment or dispatch prioritization.

Limitations

- NEMSIS is a convenience sample and is not nationally population-based (NEMSIS, 2026).
- Records are EMS activations, not unique patients or incidents; multiple agencies may submit records for one event.
- First EMS unit on scene refers to the reporting agency and may not always identify the first responder across agencies.
- Missing and not-recorded values can introduce information bias and may encode documentation practices.
- The 50,000-record source sample is a subset of the full 2025 cardiac-arrest population.
- ROSC is an intermediate outcome, not survival to discharge or favorable neurological survival.
- Unmeasured factors, including CPR quality, exact collapse-to-CPR time, comorbidities, and agency-level practices, may confound associations.
- Cross-validation estimates internal predictive performance within the sample; external validation was not performed.
- Random-forest impurity importance is descriptive and can favor continuous or high-cardinality predictors.

Conclusion

EMS system response time improved ROSC prediction in both the broad and classic Utstein-style cohorts. The incremental gain was most visible in the classic subgroup, where the slowest response-time decile also had a substantially lower ROSC rate than the fastest decile. The enhanced random forest offered the strongest threshold-independent performance, while logistic regression supplied a transparent comparison. These results support including response time in cardiac-arrest analytics while preserving careful language: the project demonstrates prediction and association, not causation.

7. Reproducibility and Data Governance

- Source archive remained outside Git and was never extracted in full.
- Preparation scripts streamed selected tables and wrote bounded local CSV files.
- The final modeling CSV omitted PcrKey and is ignored by Git.
- The executed notebook, preparation script, aggregate results, figures, and requirements file form the reproducible project record.
- NEMSIS confirmed by email that sharing the derived row-level project data with the course evaluation cohort is permitted; the repository remains private to avoid broader redistribution.

References

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Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3), e0118432. <https://doi.org/10.1371/journal.pone.0118432>

Appendix: Key NEMSIS code rules

Concept	Codes used
Resuscitation attempted	eArrest.03 contains 3003001, 3003003, or 3003005
ROSC positive	eArrest.12 contains 3012003, 3012005, or 3012007
ROSC negative	eArrest.12 contains 3012001 without a positive code
Bystander witnessed	eArrest.04 contains 3004003, 3004005, or 3004007 and not 3004001
Shockable rhythm	eArrest.11 contains 3011009, 3011011, or 3011013
Presumed cardiac	eArrest.02 = 3002001
First EMS unit	eScene.01 = 9923003
Primary emergency response	eResponse.05 = 2205001